Multi-Station Datalogger stations Quality Control analysis

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February 25, 2026

Abstract

This report presents a structured quality control and network-level assessment of 7 datalogger stations **18758,18759, 18760, 18761,18762,18763 and 18764 and the analysis was performed on data recorded since August 2023 up to February 2026**. The workflow follows meteorological data validation principles i.e timestamp validation and indexing, completeness assessment, physical plausibility checks (out-of-range), logical consistency checks (Min–Avg–Max ordering), time-gap/outage detection, stuck-sensor diagnostics (repeated values), cross-station behavior comparison (temperature and precipitation), correlation analysis, and a final overall station health ranking. The final outputs provide an operational decision-support summary for identifying healthy stations and prioritizing maintenance actions.

1 Introduction

Automatic weather stations (AWS) and datalogger sensors are widely used for continuous meteorological monitoring. However, raw sensor data can contain missing values, unrealistic measurements, inconsistent summary statistics, communication outages, and sensor malfunctions (e.g., frozen readings). Reliable downstream use (climate summaries, modeling, early warning) requires systematic QC before analysis or reporting.

1.0.1 Objective:

To evaluate station-level and network-level quality using a reproducible QC pipeline and to produce a final station health ranking for maintenance prioritization.

2 Dataset Description

The analysis used multiple station datasets, each containing time-indexed meteorological variables. Table 1 summarizes the main variables used.

Table 1: Key variables available in the station datasets.

Temperature,Average; Temperature,Minimum; Temperature,Maximum; Humidity,Avarage; Humidity,Minimum; Humidity,Maximum; Wind Speed,Average; Wind Speed,Minimum; Wind Speed,Maximum; Precipitation,Accumulation; Global solar radiation,Average

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Details of the respective datasets used for each station:
18758 (29848, 36) from 2023-08-07 16:00:00 to 2025-12-21 10:50:00
18759 (103745, 24) from 2023-08-07 16:00:00 to 2026-02-16 15:10:00
18760 (86002, 21) from 2023-08-07 16:00:00 to 2026-02-16 14:00:00
18761 (19252, 18) from 2023-08-12 16:00:00 to 2026-02-16 15:00:00
18762 (73687, 21) from 2023-08-07 16:00:00 to 2026-02-16 15:10:00
18763 (98864, 27) from 2023-02-28 17:00:00 to 2026-02-16 15:20:00
18764 (82793, 21) from 2023-08-07 16:00:00 to 2025-11-14 21:10:00
```

Figure 1: The above figure shows in summary about info of datasets used.

3 Methodology

3.1 Step 1: Import, Timestamp Parsing, and Time Indexing

Each station CSV was loaded into a pandas DataFrame. The `datetime` column (format similar to `28/02/2023 17:00`) was converted into a valid datetime object using day-first parsing. Invalid timestamps were removed, then records were sorted chronologically and indexed by time.

Let t_i denote the timestamp of observation i. Time ordering ensures:

$$t_1 < t_2 < \cdots < t_n$$

This is required for gap detection, resampling, and trend comparison.

3.2 Step 2: QC Range Thresholds

A basic physical-range evaluation was applied to detect impossible values. Table 2 shows the limits used.

Table 2: Possible Range thresholds used for out-of-range detection.

Variable	**Lower bound**	**Upper bound**
Temperature (Min/Avg/Max)	$-10°C$	$45°C$
Humidity (Min/Avg/Max)	0%	100%
Wind Speed (Min/Avg/Max)	0	60
Precipitation,Accumulation	0	500
Global solar radiation,Average	0	1400

3.3 Analysis 1: Completeness (Missingness) Comparison

Completeness measures the proportion of non-missing values. For a variable X:

$$C_X = 1 - \frac{N_{\text{missing}}}{N_{\text{total}}}$$

where C : completeness rate, and a high completeness rate (close to 1) indicates stable data availability. Lower completeness rate indicates sensor downtime or communication loss.

Interpretation

- $C > 0.99$: excellent availability
- $0.95 \leq C \leq 0.99$: acceptable, monitor
- $C < 0.95$: investigate for outages or sensor failures

Observed completeness ranking

Table 3 shows average completeness per station.

Table 3: Average completeness ranking across stations.

StationId	avg_completeness
18762	0.999993
18760	0.999962
18761	0.999912
18763	0.999740
18759	0.994799
18758	0.984586
18764	0.755158

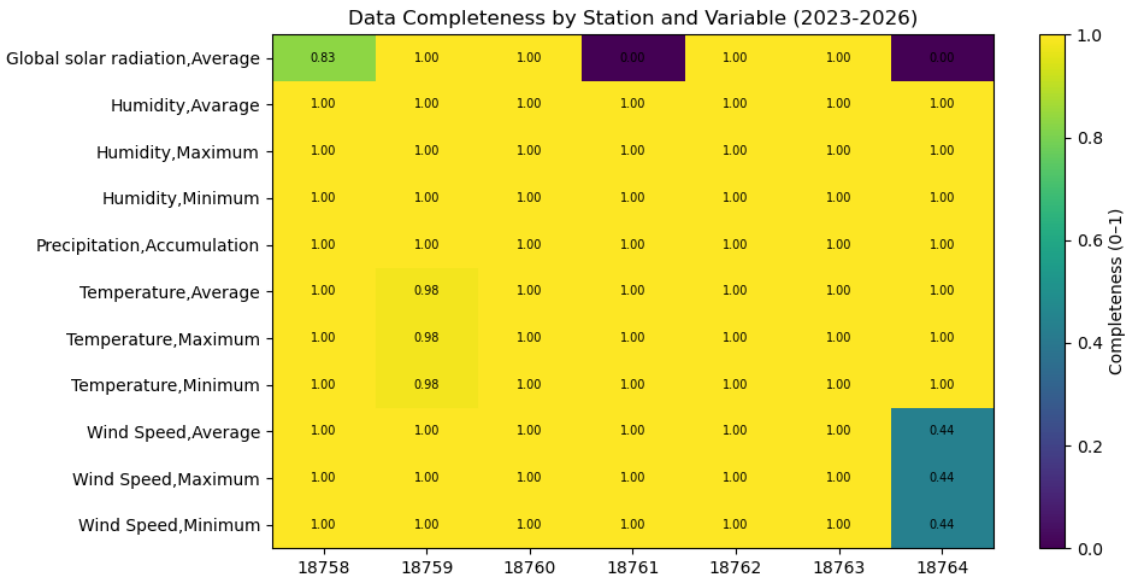

Figure 2: Completeness heatmap by station and variable.

3.4 Analysis 2: Out-of-Range + Ordering Violations (Temperature/Humidity) Respectively

3.4.1 (A) Out-of-Range Detection

Values were flagged if:

$$x < Lowerbound \quad \text{or} \quad x > Upperbound$$

Out-of-range values indicate likely sensor malfunction, unit issues, calibration problems, or decoding/mapping errors.

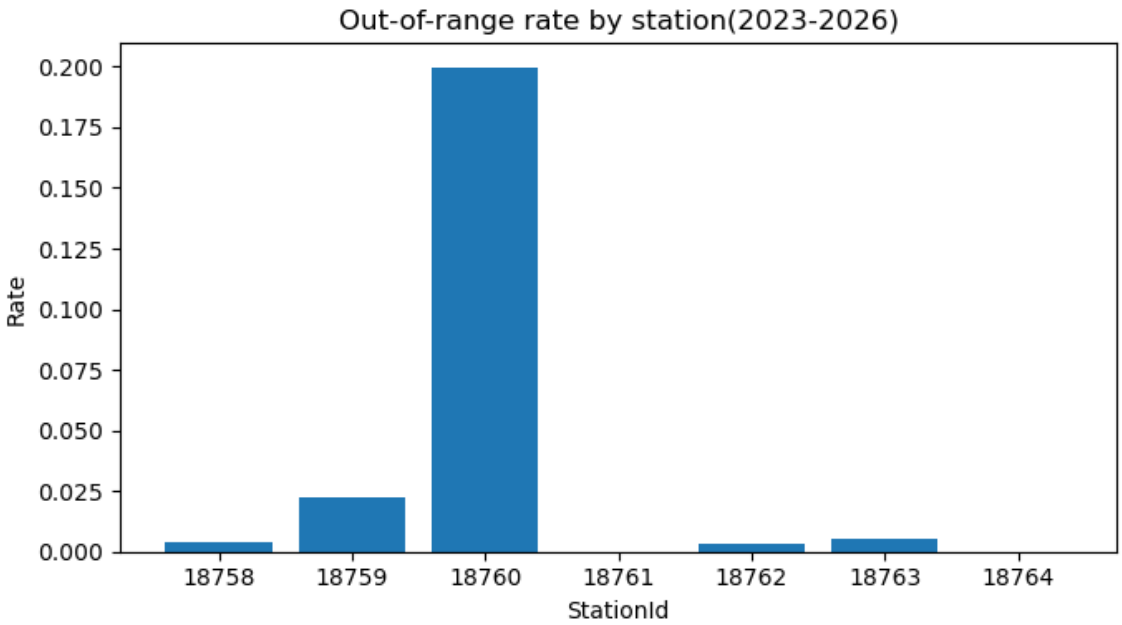

Figure 3: Out-of-range rates by station

Table 4: Out-of-Range Rate by Station from 2023-2026

Rank	StationId	Out_of_Range_Rate	Interpretation
1	18760	0.199623	Very high — requires urgent investigation
2	18759	0.022150	Moderate — should be monitored
3	18763	0.005553	Low — generally acceptable
4	18758	0.004188	Low — data quality is good
5	18762	0.002999	Very low — reliable measurements
6	18761	0.000000	Excellent — no out-of-range values
7	18764	0.000000	Excellent — no out-of-range values

3.4.2 (B) Logical Ordering Violations

For temperature and humidity, the following must hold:

$$X_{\text{min}} \leq X_{\text{avg}} \leq X_{\text{max}}$$

Violations strongly suggest channel mapping or computation errors.

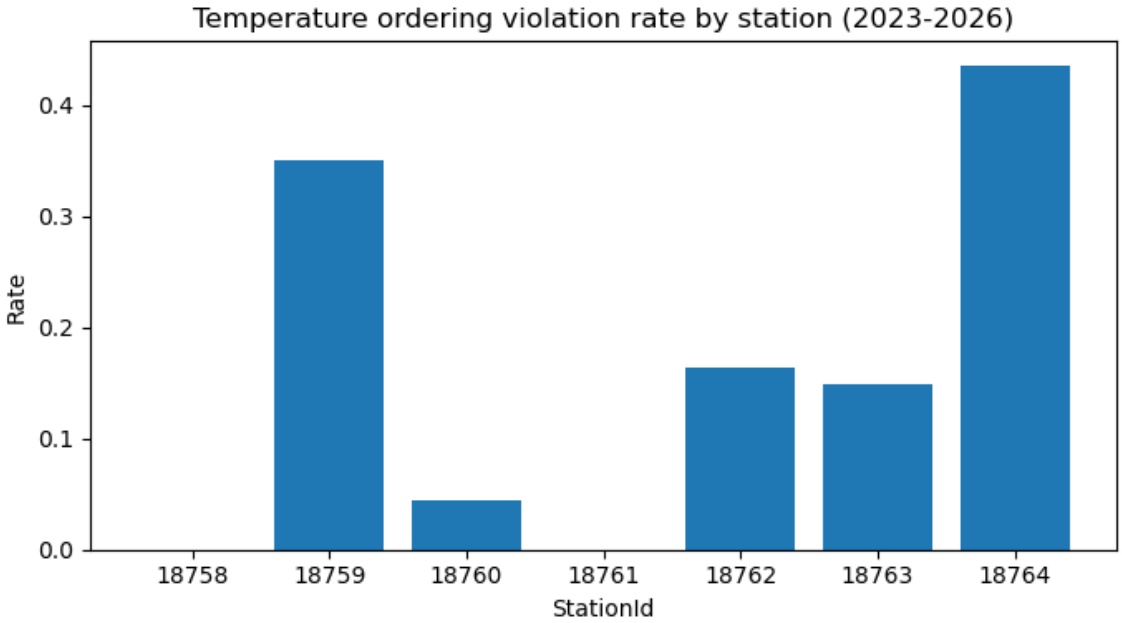

Figure 4: Temperature ordering violation rates by station.

Violation Rate Interpretation

- 0.00 : Perfect
- < 0.01 : Minor noise
- $0.01 - 0.05$: Investigate
- > 0.10 : Serious problem(likely calculation issue)

3.5 Analysis 3: Time Gaps / Outages (Data Continuity)

Time gaps between consecutive records were computed:

$$\Delta t_i = t_i - t_{i-1}$$

Large gaps indicate possible outages (power loss, network/SIM issues, logger resets, storage problems). Indicators include median gap, high-percentile gaps, and the maximum gap.

interpretation of Time gaps/outages

- $<$ 30 minutes: excellent continuity
- 30–120 minutes: acceptable
- 2–12 hours: warning
- $>$ 12 hours: serious outage
- $>$ 24 hours: critical reliability issue

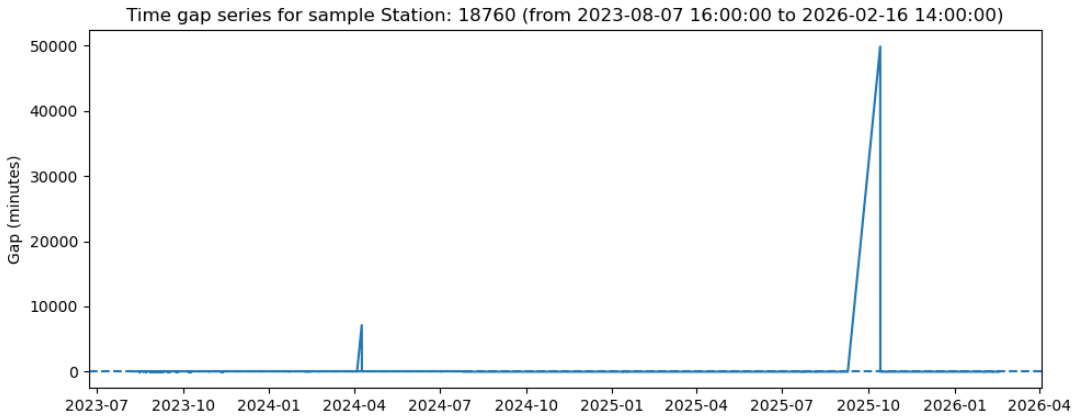

Figure 5: Maximum observed time gap on station: 18760.

3.6 Analysis 4: Stuck Sensors / Most Repeated Values (Fault Detection)

Stuck-sensor diagnostics detect sensors that repeatedly output the same value. The most common value share for variable X is:

$$S = \max_{v} \frac{\text{count}(X = v)}{N}$$

High S suggests a frozen sensor or constant-value fault.

Interpretation:

- $S < 0.8$: normal
- $0.8 \leq S < 0.95$: suspicious
- $S \geq 0.95$: likely stuck
- $S = 1.0$: almost certainly stuck

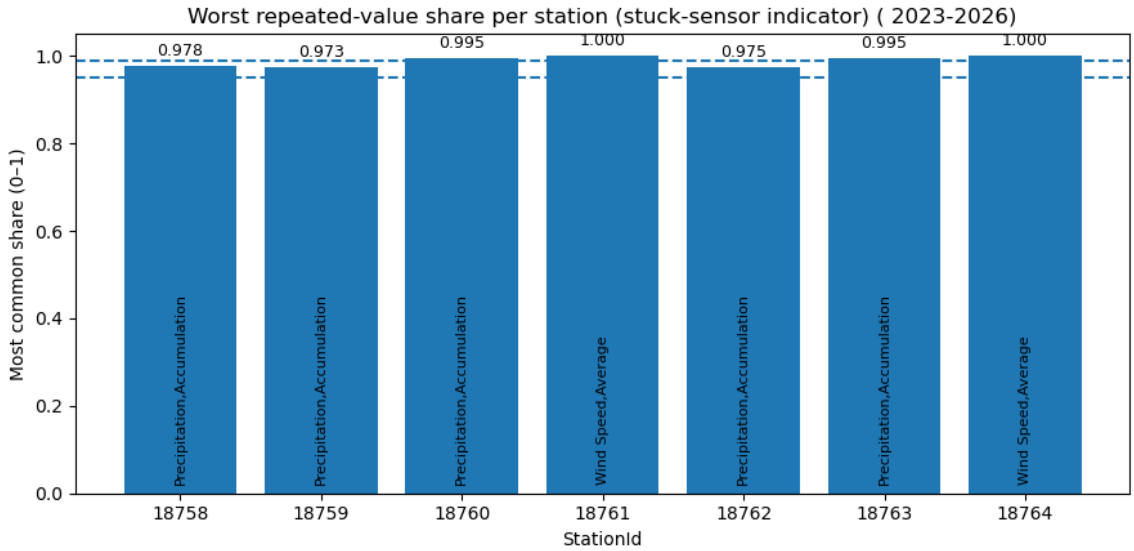

Figure 6: Worst repeated-value share per station (stuck-sensor indicator).

3.7 Analysis 5: Cross-Station Behavior (Daily Trends + Comparison)

Cross-station comparison checks whether stations behave coherently over time. Daily means were computed using resampling:

$$\bar{X}_{\text{day}} = \frac{1}{n_d} \sum_{i \in d} X_i$$

Temperature should show similar seasonal behavior across nearby stations.

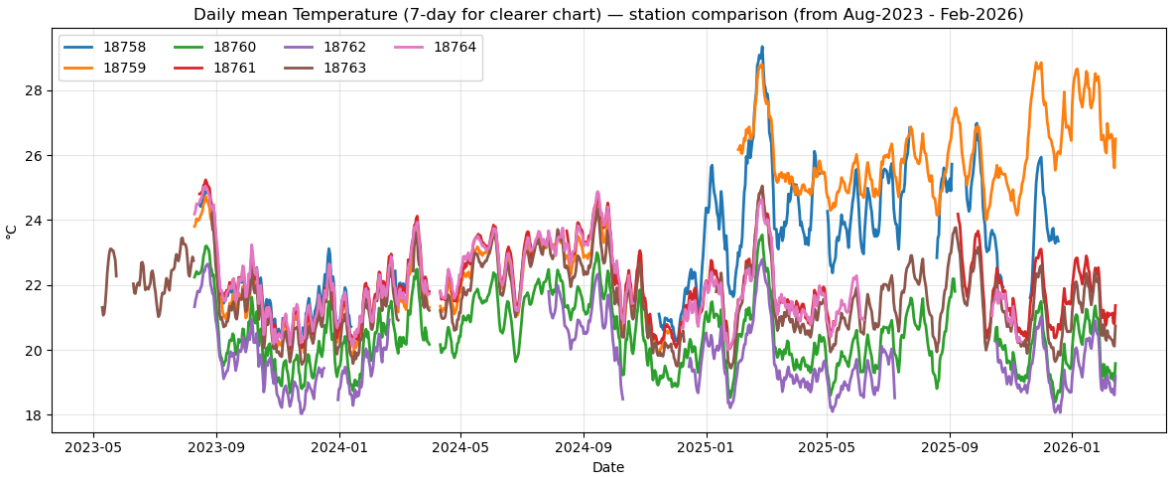

Figure 7: Daily mean temperature comparison across stations (7-day smoothed.

3.8 Rain Detection Frequency Rate

Rain detection frequency represents the proportion of days on which measurable rainfall was recorded at each station. According to World Meteorological Organization (WMO) standards, a rain day is typically defined as a day with precipitation greater than or equal to 1 mm.

The rain detection frequency is calculated as:

$$F_{\text{rain}} = \frac{N_{\text{rain days}}}{N_{\text{total days}}} \tag{1}$$

where $N_{\text{rain days}}$ is the number of days with precipitation ≥ 1 mm and $N_{\text{total days}}$ is the total number of observed days.

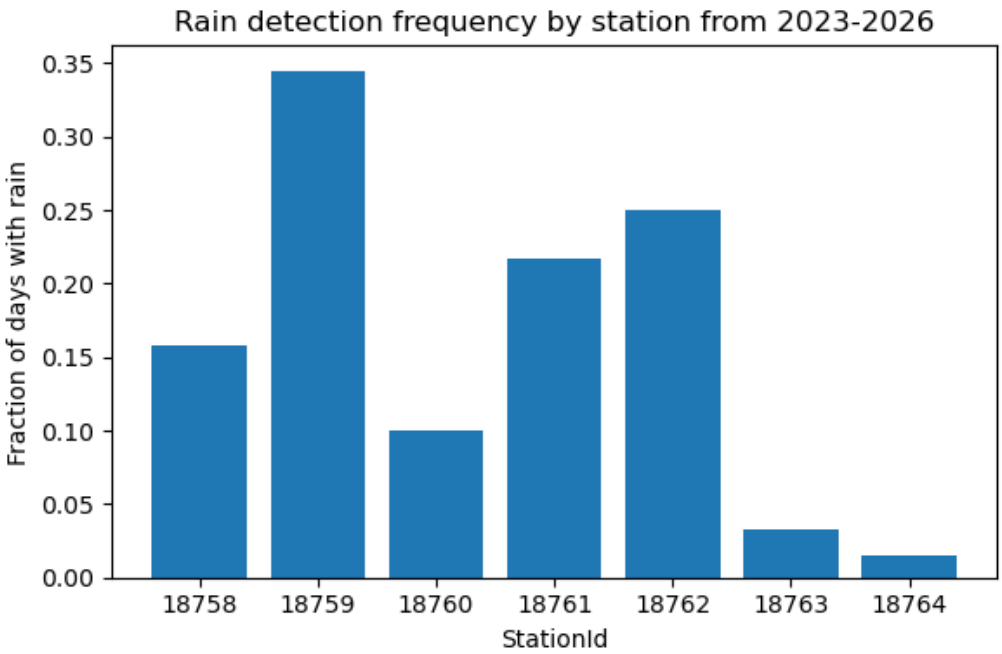

Figure 8: Rain detection frequency (fraction of days with rain) by station.

Interpretation of Results

Table 5 summarizes the rain detection frequency across stations.

Table 5: Rain Detection Frequency by Station

Rank	StationId	F_{rain}	Interpretation
1	18759	0.344700	High — relatively wet conditions
2	18762	0.249770	Moderate — climatologically reasonable
3	18761	0.216590	Moderate — climatologically reasonable
4	18758	0.158525	Low but possible
5	18760	0.100461	Low — should be monitored
6	18763	0.032258	Very low — potential data quality issue
7	18764	0.014747	Extremely low — requires investigation

Considering Rwanda's humid equatorial climate under WMO allignment, the extremely low rain detection frequencies observed at Stations 18763 and 18764 may indicate rain gauge under-catch, blockage, calibration problems, or data gaps. These stations should therefore be prioritized for detailed quality control.

Overall, most stations show rainfall frequencies consistent with Rwanda's tropical climate variability. However, Stations **18763 and 18764** exhibit unusual low rain occurrence and should be flagged for further technical verification.

3.9 Correlation Analysis (Temperature Cross-Station)

Correlation measures whether stations rise/fall together. Pearson correlation between stations i and j:

$$\rho_{ij} = \text{corr}(X_i, X_j)$$

High ρ indicates strong agreement and network coherence. Persistent low correlation may indicate calibration drift, site differences, or measurement problems.

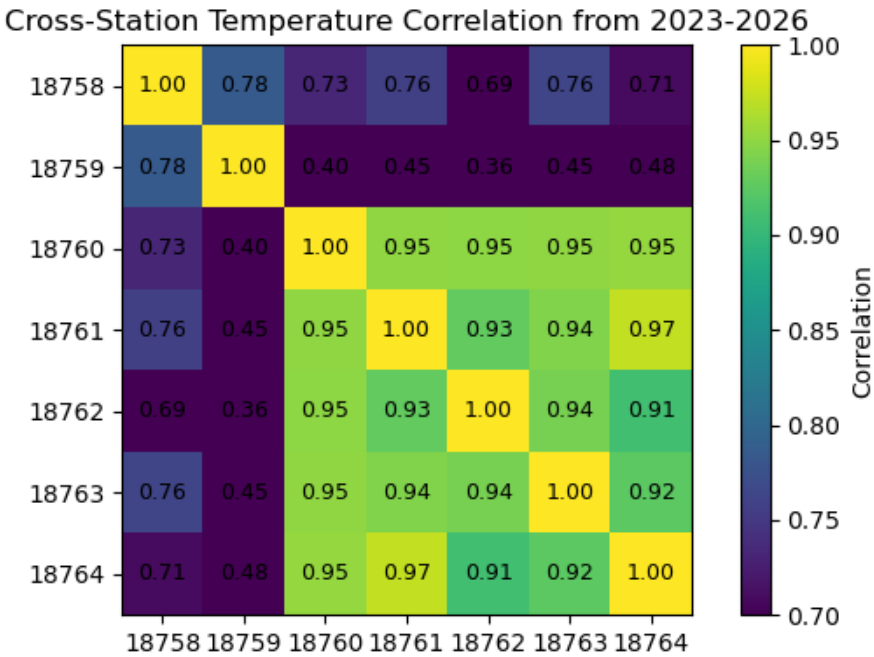

Figure 9: Cross-station temperature correlation matrix.

4 Final overall Station Health Ranking

4.1 Health Score Definition

An overall station health score was computed as:

$$H = 1 - (0.30R + 0.30O + 0.15M + 0.15G + 0.10S)$$

where:

- R = out-of-range rate (physical implausibility)
- O = ordering violation rate (logical inconsistency)
- M = missingness penalty ($1 -$ completeness)
- G = normalized gap severity (outage indicator)
- S = worst repeated-value share (stuck-sensor indicator)

Meaning: The score starts at 1 (perfect station) and subtracts weighted penalties for QC issues. Higher H is better.

4.2 Observed Health Ranking

Table 6 summarizes final station scores and key QC indicators.

Table 6: Final station health ranking and contributing QC indicators.

StationId	Health	Comp.	OOR	Order	Max gap (min)	Stuck share
18761	0.758639	0.999912	0.000000	0.000000	168720.0	1.000000
18760	0.754757	0.999962	0.018149	0.044162	49820.0	0.995221
18758	0.749745	0.984586	0.000458	0.000000	352490.0	0.978052
18763	0.720832	0.999740	0.000505	0.149549	94830.0	0.995317
18762	0.708730	0.999993	0.000273	0.163964	220300.0	0.975178
18759	0.672320	0.994799	0.002014	0.349790	38737.0	0.972914
18764	0.590818	0.755158	0.000000	0.435484	175510.0	1.000000

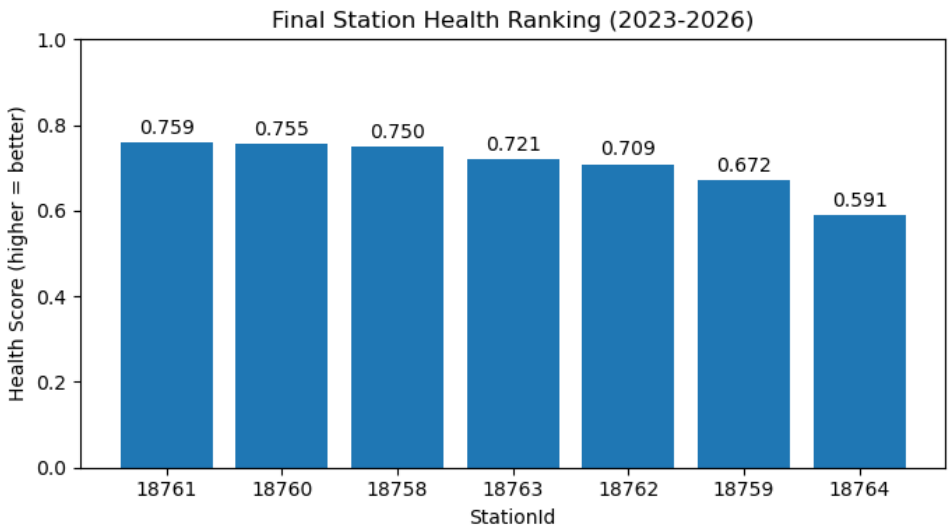

Figure 10: Final overall station health ranking (higher is better).

4.2.1 Interpretation of the Overall station health(Score Meaning)

- > 0.90 :excellent
- 0.80—0.90 : good
- 0.70–0.80 :monitor
- 0.60–0.70 :warning
- < 0.60 :needs maintenance

5 Discussion and Maintenance Recommendations

5.1 Key Findings

- Most stations exhibit excellent overall completeness (near 1.0), indicating stable data acquisition.

- Logical ordering violations are the most critical issue in the network for certain stations (high O), suggesting potential channel mapping or computation inconsistencies.
- Several stations show large maximum time gaps, indicating outages that reduce continuity.
- Stuck-sensor indicators are high for some stations and should be validated by checking which variable is constant (e.g., wind speed constant at zero as observed in some stations like **18761** and **18764**).
- Station 18764 shows the lowest overall health score, driven by moderate completeness, high ordering violations, large gaps, and strong stuck-sensor signals.

5.2 Recommended Actions

- **Immediate investigation:** 18764 (multiple QC issues, lowest health score).
- **High priority check:** 18759 (very high ordering violation rate suggests mapping/calculation issue).
- **Continuity improvement:** stations with very large max gaps (check power/network stability, logger resets).
- **Sensor-specific checks:** stations with stuck-sensor share close to 1.0 (inspect wind/rain sensors and channel wiring).

6 Conclusion

A complete QC workflow was implemented from raw time parsing to final composite health ranking. The station health score provides an operational decision-support indicator for identifying reliable stations and prioritizing maintenance. Overall, the network appears generally operational, but specific stations require targeted corrective actions to improve logical consistency, continuity, and sensor reliability.

IMPORTANT NOTE:I used python in the analysis